

Dwaar (دوّار)

Circular-Economy Logistics Platform

حوّل النفايات إلى قيمة — *Turn waste into value*

Project Documentation & Progress Summary

Arabic-first waste-recycling app for Jordan

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1. Executive Summary

Dwaar (دوّار) is an Arabic-first, RTL mobile application that organises the collection and recycling of waste materials across Jordan (Amman, Zarqa, Irbid, Aqaba). It connects three parties in a single circular-economy loop:

- Suppliers (individuals, stores and restaurants) who post pickup requests for recyclable materials — for free.
- Drivers who accept those jobs and earn income per trip.
- Recycling companies that receive organised shipments and post their own collection jobs.

The platform tracks economic value created and waste diverted from landfills. The app is built in Flutter using an MVVM + Provider architecture, with a hybrid data layer: an in-memory/local store backs the live UI today while a Supabase backend (auth, storage, realtime, Postgres + PostGIS) is being progressively wired in.

Current phase: MVP feature-complete on mock/local data, with Supabase infrastructure written and partially connected. Recent work delivered an order-chat system, a redesigned cross-role profile with government payment (eFawateercom), and improved order tracking for suppliers and restaurants.

2. Product Overview

2.1 The Problem

- Improper disposal of used oil and recyclable materials across Jordan.
- No structured system connecting suppliers to recycling drivers.
- Lost economic value from recyclable materials and few green jobs for drivers.
- Recycling companies lack a reliable intake pipeline.

2.2 The Solution Flow

Supplier posts a request → Driver accepts → Materials are collected → Recycling company processes → Environmental and economic impact is tracked.

2.3 User Roles

Role	Arabic	Responsibility
Supplier — Individual	مورد فردي	Posts pickup requests from home
Supplier — Store / Restaurant	مورد متجر / مطعم	Posts bulk pickup requests
Driver	سائق	Accepts pickup jobs, earns per trip
Recycling Company	شركة إعادة تدوير	Receives shipments, posts collection jobs

3. Technology & Architecture

3.1 Stack

Layer	Technology
Framework	Flutter 3.x (Dart SDK ^3.11)
State management	Provider + ChangeNotifier ViewModels
Architecture	Feature-scoped MVVM
Navigation	Imperative Navigator (MaterialPageRoute)
Backend	Supabase — Postgres, PostGIS, Auth, Storage, Realtime
Models	freezed + json_serializable (Order, OrderProof)
AI	Google Gemini (document scan, marketplace assist)
Fonts	Cairo (Arabic) + DM Sans (numerals) via google_fonts
Localisation	Bilingual ar/en via .arb files (Arabic template)
Maps / location	geolocator, geocoding, PostGIS proximity

3.2 Layering

The codebase separates concerns into four layers under lib/:

- **domain/** — interfaces only (i_*_repository, i_*_service), entities, AppFailure.
- **data/** — implementations: mock_* and supabase_* repositories, services, freezed models.
- **core/** — config, constants, theme, Result type, ViewState.
- **ui/** — feature-scoped MVVM (auth, chat, chatbot, earnings, home, splash).

3.3 Data Flow

Data flow is hybrid. AppOrderStore is the central ChangeNotifier for all order state, provided app-wide. It writes to a local JSON store (survives restarts) and mirrors to SupabaseOrderRepository. Mock and Supabase implementations are bound through domain interfaces in main.dart's MultiProvider, so swapping mock → real is a one-line change.

3.4 Design System

Token	Hex	Usage
Primary Green	#0F5A34	Headers, active states, primary buttons
Primary Dark	#06331C	Gradient ends, high-contrast surfaces
Accent Amber	#D97706	Points, pending status, earnings
eFawateercom Teal	#00796B	Government payment row (CBJ brand)
Background	#F3F7F5	Screen backgrounds
Muted Text	#6A7973	Secondary labels

Light and dark themes are token-driven (AppColors), toggled by AppThemeNotifier. The whole app is forced portrait and defaults to RTL.

4. Features Delivered

4.1 Onboarding & Authentication

- Animated splash screen → login (phone or email) → OTP verification → role-aware home.
- Phone-first, progressive, role-aware signup wizard with AI document scanning (vehicle registration, business licence) via Gemini.
- MockAuthRepository today; SupabaseAuthRepository written and partially wired (phone OTP).

4.2 Role Home Shells

Each role has its own home shell with a bottom navigation bar and feature-scoped ViewModels:

- Driver — Home (available jobs, availability toggle, KPIs), Orders, Profile; live GPS publishing, proximity/arrival verification, ghost-job timers, earnings & wallet.
- Supplier (individual) & Restaurant — Home, Marketplace, My Orders, Profile; pickup-request wizard.
- Recycling Company — Home, Orders, Profile; posts collection jobs, tracks shipments & processed weight.

4.3 Orders & Marketplace

- Order is a freezed model with rich enums: OrderType, OrderStatus, 15 WasteTypes, WasteForm, WeightCategory, PickupTarget, VehicleType.
- Order lifecycle: pending → accepted → arrivedAtPickup → inTransit → arrivedAtDropoff → completed (or cancelled).
- Shared marketplace tab across all roles, role-aware (driver claims, supplier purchases, company receives).
- Full Order Details view: status timeline, route/map section, driver & customer cards, proof upload, earnings, action buttons.
- Collection-sale lifecycle for company-posted jobs with per-kg payment and weight confirmation.

4.4 Pricing, Wallet & Rewards

A fee calculator prices each job from a base fee, distance, weight surcharge, material rate and urgency bonus, then applies a platform commission. Drivers have an escrow-style wallet (available balance + held amount); suppliers earn reward points.

Component	Basis
Base fee	Vehicle-type dependent
Distance fee	km × per-vehicle rate
Weight surcharge	Tiered by total kg
Material fee	kg × per-material rate
Urgency bonus	Flat add-on for urgent jobs
Platform cut	Commission on gross fee

4.5 Order Chat (recent)

- Order-scoped 1:1 chat between the customer and the assigned driver, one room per order.
- Pinned order-context card, state-aware quick replies, and an AppBar subtitle showing the other party + live order status.
- Attachments: image from gallery and live GPS location sharing; BiDi-aware message bubbles.
- Optimistic send with retry on failure, a typing indicator via Supabase Realtime Broadcast, and a room-lock that disables sending once an order is completed/cancelled while keeping history readable.

4.6 Profile Redesign + Government Payment (recent)

- Unified, modern profile shared across all three roles: gradient hero, floating stats card, info sections, app settings, help, and account actions.
- Six reusable profile widgets extracted, eliminating ~70% duplicated code across the three old profile tabs.
- **New payment section** with a role-aware card — Driver: wallet balance + held funds + withdraw; Supplier: points + progress to next reward; Company: billing-period summary.
- **eFawateercom (فواتيركم)** integration row — Jordan's national e-payment gateway operated by the Central Bank of Jordan — the correct rail for a government-adjacent service.
- Fixed dark-mode defects and removed duplicate edit actions in the process.

4.7 Order Tracking Improvements (recent)

- Upgraded the shared progress stepper to distinguish completed / current / upcoming steps with a 'you are here' halo — benefiting driver, supplier and restaurant cards.
- Added a plain-language live status banner with an ETA pill on active supplier/restaurant orders, plus a clear 'Track order' (تتبع الطلب) button.

4.8 AI Capabilities

- Gemini-powered scanning of vehicle registration and business licences to auto-fill signup fields.
- Marketplace assistant and asset-scan flows (e.g. oil/wood) with quick-reply chips and scanning loaders.
- All AI calls sit behind service interfaces with mock implementations for offline/test paths.

5. Backend & Security

The Supabase backend is defined through versioned SQL migrations covering schema, chat, PostGIS compatibility, chat media, security hardening, RLS fixes, driver locations, marketplace limits, transaction commission, vehicle types and wallet functions.

5.1 Key Backend Capabilities

- Postgres schema for users, orders, chat and wallets, with Row-Level Security policies scoping data to its participants.
- PostGIS-backed driver location streaming and server-side proximity/arrival verification (anti-fraud).
- Wallet hold/release RPCs for order escrow and a daily payout edge function.

- Private chat-attachments storage bucket (images only, size-limited) with participant-scoped access.

5.2 Security Posture

- Security hardening migrations applied: scoped RLS, revoked public execute, reset PostGIS search path, patched auth.uid() usage.
- **Open item:** mockAuth must be set to false in main.dart before production so Supabase phone OTP is enforced.

6. Status & Roadmap

6.1 Done

- Onboarding, login, OTP, role-aware signup wizard with AI scan
- All three role home shells with feature-scoped ViewModels
- Order model, full lifecycle, shared marketplace and order details
- Pricing engine, driver wallet, supplier rewards
- Order-scoped chat with media, optimistic send and typing indicator
- Redesigned cross-role profile with eFawateercom government payment
- Improved order tracking (stepper + live status banner + track CTA)
- Supabase infrastructure (migrations, RLS, PostGIS, wallet, storage)

6.2 Next

- Flip mockAuth → false and complete live Supabase auth wiring
- Push notifications (FCM) for orders and chat
- Unread chat badges on order cards
- Real-time driver tracking on the customer side + map polish
- Payout flow and eFawateercom biller registration
- Chat localisation + pagination; future: auto-translate, voice messages

Dwaar (دَوَّار) — internal project documentation. Generated 22 June 2026.